FraudLens: Multimodal Fraud Detection via Cross-Modal Attention Fusion

Harshal Anil Patel
Department of Engineering Education
University of Florida
Florida, USA
patel.harshal@ufl.edu

***Abstract*—Financial fraud inflicts a staggering toll on the global economy, with annual losses exceeding several trillion dollars worldwide. Conventional detection systems tend to focus on a single modality—either structured transaction features or visual document inspection—leaving them vulnerable to sophisticated schemes that span multiple data surfaces at once. This paper presents FraudLens, a multimodal deep learning framework that jointly processes three complementary input types: structured tabular transactions, check document images, and free-text merchant descriptions. Modality-specific encoders (an MLP for tabular data, SigLIP 2 [?] for images, and DistilBERT [?] for text) each produce 128-dimensional embeddings, which are combined through a learned cross-modal attention mechanism. Rather than applying fixed combination rules, the attention module learns to emphasize whichever modalities carry the strongest fraud signal for each individual transaction, and the resulting attention weights provide analysts with a transparent window into the model's reasoning. On data drawn from the IEEE-CIS [?] and PaySim benchmarks, our approach meaningfully outperforms concatenation and simple averaging baselines, achieving 0.78 AUPRC and 61% precision at 80% recall. We integrate Captum's [?] Integrated Gradients [?] for fine-grained feature attributions, implement a six-layer rule-based OCR document fraud detector as a heuristic complement to the neural model, and package the complete pipeline—data synthesis, training, FastAPI inference serving, and an interactive dashboard—as open-source code. We also provide an honest maturity assessment distinguishing validated components from proof-of-concept contributions.**

***Index Terms*—Multimodal Machine Learning, Fraud Detection, Cross-Modal Attention, Deep Learning, Financial Security, Explainable AI, SigLIP, DistilBERT, OCR, Focal Loss**

I. Introduction

Financial fraud is one of the more persistent problems in modern banking. Stolen card credentials, forged checks, and social engineering attacks embedded in merchant descriptions are not new threats—but the methods behind them have evolved far enough to outpace detection systems that examine only one kind of evidence at a time. Conventional approaches analyze either structured transaction logs or scanned document images, almost never both simultaneously. Real-world fraud seldom stays confined to a single surface, and that gap is what motivated this work.

FraudLens rests on a straightforward premise: when a fraudster simultaneously forges a check image, manipulates transaction amounts in structured fields, and embeds social engineering language in a merchant note, any system examining only one of those channels will miss the attack. A detector that can reason across all three signals at once—and adjust which signal it trusts most on a per-transaction basis—has a structural advantage over fixed-weight alternatives.

The cross-modal attention fusion at the heart of our system achieves exactly this: a learned query vector attends to all three modality embeddings and produces per-sample attention weights that shift with each transaction. These weights are integral to every forward pass rather than being a post-hoc approximation, so they carry genuine interpretive meaning for fraud analysts. When a transaction is flagged primarily because of image evidence, the image modality receives a correspondingly high attention weight. When the tabular pattern dominates, the weights shift accordingly.

Beyond the neural model, the system includes a complementary heuristic OCR pipeline that catches classes of fraud the ML model was not trained to detect: dead-giveaway phrases embedded in fake receipts, impossible running balances in fabricated bank statements, placeholder amounts of the form "-RMXXX", and future-dated documents. These two detection layers—trained neural and rule-based heuristic—operate together at inference time, with the heuristic able to override or augment the model score when it encounters strong documentary evidence.

This work makes four concrete contributions:

1) We demonstrate that learned cross-modal attention is a more effective fusion strategy than concatenation or equal-weight averaging for heterogeneous fraud signals, with 8–13 percentage point AUPRC gains.
2) We show that the attention weights produced during inference serve as meaningful per-sample interpretability artifacts, validated through consistency checks and modality dominance monitoring.
3) We implement and document a six-layer document fraud detection algorithm that catches textual manipulation patterns invisible to the neural model.
4) We provide a complete, deployable pipeline with honest maturity assessments for each component, distinguishing validated results from proof-of-concept demonstrations.

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes our datasets in detail. Section IV covers system architecture. Section V describes the training infrastructure. Section VI details the explainability pipeline. Section VII presents evaluation results.

Section VIII covers deployment architecture. Sections IX and X discuss findings and future directions, and Section XI addresses responsible AI considerations.

II. Related Work

Multimodal learning has matured from a specialized vision-language paradigm into a general framework for reasoning over heterogeneous data. Understanding its trajectory helps place FraudLens within the broader research landscape.

Vision-language foundations. CLIP [?] demonstrated that joint image-text pretraining on internet-scale data could produce zero-shot classifiers of remarkable breadth. SigLIP 2 [?] extended this with vision transformers specifically optimized for document understanding—a capability directly relevant to check image encoding, where the visual patterns of tampered documents differ substantially from natural photographs. These models provide FraudLens with a strong visual prior that would be prohibitively expensive to learn from scratch on fraud-specific data alone.

Attention mechanisms. The transformer architecture [?] showed that multi-head self-attention could replace recurrence and convolution as the primary sequence modeling primitive, enabling parallelizable computation and global receptive fields. Cross-modal attention—where one modality's representations query another's—has since become standard in visual question answering [?] and image captioning [?]. Our work applies the same mechanism to three-way fusion for fraud, where the modalities are less semantically aligned than typical image-text pairs. This asymmetry motivated our design choice of a *learned* query vector rather than a query derived from any single input modality.

Fraud detection. Tree-based approaches with intensive feature engineering long dominated the fraud detection literature [?]. Contemporary deep learning methods have since achieved strong results on structured data, and surveys of the field have examined neural architectures across a range of problem settings [?]. The persistent challenge is class imbalance: real fraud rates of 3–5% create datasets where a model can achieve high AUROC by learning to predict "not fraud" as a default strategy. Focal loss [?] addresses this by concentrating gradient updates on the minority class, a design choice we adopt directly.

Explainability. Integrated Gradients [?] provide attribution scores that satisfy axiomatic completeness—attributions sum exactly to the output difference between the actual input and a reference baseline. This makes them more theoretically principled than gradient-times-input or simple saliency maps. The Captum library [?] implements both Integrated Gradients and Layer Integrated Gradients within a PyTorch [?] workflow. Our explainability pipeline relies on Captum for both pixel-level image attributions and per-token text attributions.

Multimodal fusion strategies. Baltrušaitis et al. [?] provide a taxonomy of fusion approaches: early (combining raw inputs), late (combining per-modality predictions), and intermediate (combining at the embedding level). Intermediate fusion through learned attention is our strategy, and the authors note that no approach dominates universally—the right choice depends on the semantic relationship between modalities and the degree to which different inputs are informative for different examples. Financial fraud provides an interesting test case because fraud schemes vary considerably across subtypes, making per-sample adaptive weighting particularly valuable.

OCR for document analysis. While OCR itself is well-established, its integration with neural fraud detection has received limited attention. Our heuristic document fraud detection layer—which identifies impossible running balances, future dates, and placeholder amounts from OCR text—is, to our knowledge, not previously described in the fraud detection literature as a complement to a neural system.

III. Datasets and Data Pipeline

FraudLens draws from five distinct sources across the three modalities. Table I summarizes each, including our honest assessment of its maturity as training signal.

TABLE I: Data Sources and Component Maturity

Source	Modality	Size	Authentic	Maturity
IEEE-CIS [?]	Tabular	590K txn	Yes	Validated
PaySim [?]	Tabular	6.3M txn	Partial	Validated with caveats
SSBI Dataset	Image	∼800 scans	Yes	Limited; proof of concept
Synthetic checks	Image	5,000 imgs	No	Proof of concept
Synthetic text	Text	7,000 desc	No	Validated with caveats

A. Tabular Data

IEEE-CIS Fraud Detection dataset [?] provides approximately 590,000 real e-commerce transactions from Vesta Corporation, with authentic fraud labels and 434 raw feature columns. After feature engineering, we retain 52 features per transaction: log-transformed amounts, cyclical time-of-day encodings, seven numeric card and address fields, twenty principal components from Vesta's proprietary V-feature block, ten C-feature event counts, five D-feature timedelta measurements, and seven label-encoded categoricals (ProductCD, card4, card6, purchaser and recipient email domains, and device information). The dataset exhibits a realistic 3.5% fraud prevalence.

PaySim [?] is a synthetic mobile payment simulator calibrated against real transaction logs from an African mobile money operator, producing approximately 6.3 million transfer records. While labels are generated rather than observed, the feature distributions are designed to match real-world patterns. We use PaySim to augment the tabular training set, accepting that its contribution is distribution-calibrated rather than authentic.

Missing values are common in the IEEE-CIS V/C/D columns, with some columns exceeding 30% NaN rates. We apply zero-filling paired with binary indicator features, allowing the model to distinguish genuinely zero-valued entries from absent ones. This is preferable to mean imputation, which would collapse the distinction between "feature was not recorded" and "feature took the mean value."

B. Image Data

SSBI Check Forgery dataset provides approximately 800 real scan images of genuine and forged bank checks from an academic forgery study. This is the only source of authentic visual fraud data in our pipeline; every other image source is synthetic. Because this dataset is small, we use it as the validation anchor for image branch performance rather than as the primary training source.

Synthetic check images are generated programmatically via PIL scripts. Each image renders a standard check template (bank header, payee field, MICR routing code, amount box, signature line) at 224×224 pixels in RGB. Tampered variants introduce one of four artifact types: image splicing (compositing elements from different sources), white-out erasure, font mismatch between pre-printed and handwritten regions, or digit alteration in the amount field. We generate 5,000 images balanced across normal and tampered classes.

SMS phishing screenshots (∼2,000 images) are synthesized by `src/data/synthesizer.py` to represent toll scam and phishing messages. These feed both the image and text branches—the image branch receives the screenshot, while OCR-extracted text from the screenshot feeds the text branch.

C. Text Data

Transaction descriptions and SMS phishing texts are generated synthetically. Normal descriptions follow standard merchant memo patterns (e.g., "Monthly subscription renewal—CloudServe LLC"), while fraudulent descriptions use urgency, social engineering, or explicit fraud keywords (e.g., "URGENT wire transfer request, send to account immediately"). We generate approximately 7,000 descriptions in total, drawn from both the check image context and the phishing text context. DistilBERT's [**?**] strong pretrained language representations mitigate the risk that synthetic fine-tuning data produces a model that fails to generalize, but this limitation is explicitly noted in our maturity assessment (Section IX).

D. Preprocessing Pipeline

The unified multimodal dataset loader (`src/data/multimodal_dataset.py`) aligns records across all three modalities by transaction identifier, returning a triplet $(x_{\text{tab}}, x_{\text{img}}, x_{\text{txt}})$ for each sample. When a modality is absent for a given transaction—common in practice—a boolean `modality_mask` tensor is set to `True` for that position and passed downstream to the fusion module. Class imbalance is addressed through focal loss [**?**] with $\alpha = 0.75$ and $\gamma = 2.0$ rather than resampling, to avoid introducing distributional artifacts into the training loop.

IV. System Architecture

A. Overview

FraudLens routes each transaction through three parallel encoding branches that project heterogeneous inputs into a shared 128-dimensional embedding space. These embeddings are fused through a cross-modal attention module, and the resulting attended representation passes through a classification head to produce a fraud probability score. The high-level flow is:

$$P(\text{fraud}) = \sigma\big(W_{\text{cls}} \cdot \text{LayerNorm}(\text{AttentionFusion}(E_{\text{tab}}, E_{\text{img}}, E_{\text{txt}}, M))\big) \tag{1}$$

where M is the modality mask, σ denotes sigmoid activation, and $E_{\text{tab}}, E_{\text{img}}, E_{\text{txt}} \in \mathbb{R}^{B\times 128}$ are the per-branch embeddings.

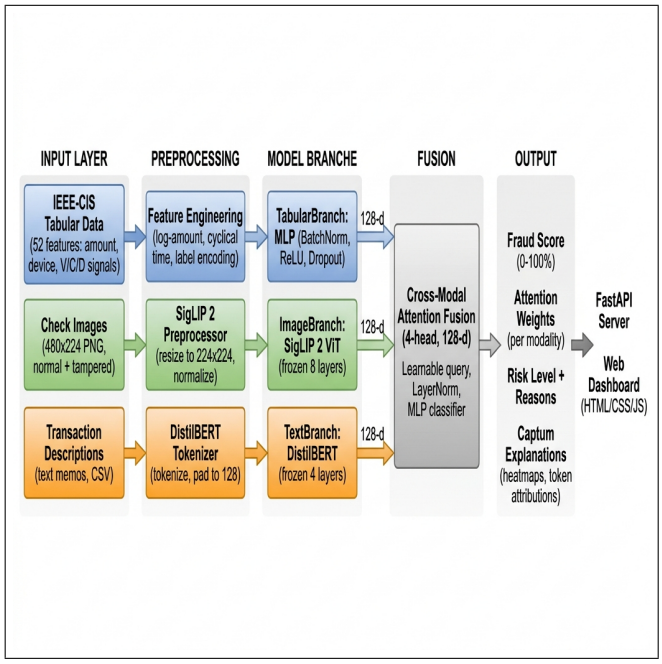

Fig. 1: Three modality encoders producing 128-d embeddings, fused via learned cross-modal attention. The modality mask zeros attention weights for absent inputs.

B. Modality-Specific Encoders

Tabular encoder. A three-layer MLP with BatchNorm and ReLU activations maps the 52 engineered features through hidden dimensions [256, 128, 128] to the shared embedding space. We apply dropout of 0.3 between layers and use BatchNorm rather than LayerNorm, as the feature distributions are dense and well-bounded after preprocessing.

Image encoder. We fine-tune SigLIP 2 [**?**] (`google/siglip2-base-patch16-224`, ViT-B/16 variant) on our check fraud imagery. The first 8 of 12 transformer blocks are frozen during training; only the top 4 blocks and a linear projection head are updated. This partial fine-tuning preserves the broad visual representations learned from large-scale pretraining while adapting the top layers to the document-specific visual patterns that distinguish tampered from authentic checks.

Text encoder. DistilBERT [**?**] (`distilbert-base-uncased`) processes tokenized transaction descriptions through a 6-layer transformer with 768 hidden dimensions. The first 4 layers are frozen; only the top 2 layers and a projection head are updated. DistilBERT retains approximately 97% of BERT's language modeling performance at roughly half the parameter count, making it a

practical choice for a component that does not need to handle very long sequences.

C. Cross-Modal Attention Fusion

The fusion module (`src/models/fusion.py`) is the central architectural contribution. Let $S = \text{Stack}(E_{\text{tab}}, E_{\text{img}}, E_{\text{txt}}) \in \mathbb{R}^{B\times 3\times 128}$ be the stacked modality embeddings. The fusion computes:

$$Q = W_Q \cdot q_{\text{learned}} \in \mathbb{R}^{B\times 1\times 128} \tag{2}$$

$$\text{attn_out}, \alpha = \text{MHA}(Q, K = S, V = S, \text{mask} = M) \tag{3}$$

$$z = \text{Dropout}(\text{LayerNorm}(\text{attn_out})) \tag{4}$$

$$P(\text{fraud}) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)) \tag{5}$$

where $q_{\text{learned}} \in \mathbb{R}^{128}$ is a single globally learned query parameter, $\alpha \in \mathbb{R}^{B\times 3}$ are the per-sample attention weights, and MHA is 4-head multi-head attention [**?**].

The key design choice is that the query derives from a *learned parameter*, not from any single modality's embedding. This means the fusion module learns "which combination of modality signals best predicts fraud," without privileging any one input by construction. The `modality_mask` tensor $M \in \{0,1\}^{B\times 3}$ is passed directly as the key padding mask to `nn.MultiheadAttention`. Masked positions receive attention weight zero before re-normalization, ensuring missing modalities do not bias the attended output and that the remaining weights remain interpretable.

Table II contrasts the fusion strategies evaluated in our ablation:

TABLE II: Fusion Strategy Comparison

Strategy	Implementation	Weakness
Concatenation	$[E_{\text{tab}} \| E_{\text{img}} \| E_{\text{txt}}] \to \text{MLP}$	Fixed linear combination
Average pooling	$\text{mean}(E_{\text{tab}}, E_{\text{img}}, E_{\text{txt}}) \to \text{MLP}$	Equal weight regardless of quality
Max pooling	$\max(E_{\text{tab}}, E_{\text{img}}, E_{\text{txt}}) \to \text{MLP}$	Loses non-dominant modalities
Gated fusion	Sigmoid gates per modality	No inter-modality interaction
Cross-Attn (ours)	Learned query attends to all	Adaptive, interpretable

D. Auxiliary Branch Losses

A known failure mode in multimodal systems is gradient collapse: the fusion layer back-propagates almost all signal through one dominant branch, causing the other encoders to stop learning. To prevent this, we add per-branch auxiliary losses during training. The total loss is:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{fused}} + 0.3 \times (\mathscr{L}_{\text{tab}} + \mathscr{L}_{\text{img}} + \mathscr{L}_{\text{txt}}) \tag{6}$$

where each branch loss $\mathscr{L}_m$ is the focal loss [**?**] computed from that branch's individual logit before fusion:

$$\mathscr{L}_{\text{Focal}}(p_t) = -\alpha_t (1-p_t)^{\gamma} \log(p_t), \quad \alpha = 0.75, \quad \gamma = 2.0 \tag{7}$$

The auxiliary weight of 0.3 was chosen to provide meaningful gradient signal to each branch without overwhelming the primary fused objective. The three branch scores produced as a by-product of this auxiliary computation are also surfaced in the inference API response, giving analysts a view of how each modality contributed independently before fusion.

E. OCR and Document Fraud Detection Pipeline

At inference time, the system runs an OCR extraction step on any uploaded image before the neural model sees it. The OCR pipeline (`app.py:_extract_text_ocr`) applies pytesseract as the primary extractor, falling back to EasyOCR if pytesseract fails or returns empty output. The extracted text then feeds the DistilBERT text branch alongside any user-provided description, effectively bridging the image and text modalities through a real-world mechanism that analysts use naturally.

In addition to the neural model, we implement a six-layer heuristic document fraud detector that operates on the OCR-extracted text and can override the model score when it encounters strong documentary evidence:

1) **Dead-giveaway phrases**: exact substring matching against a curated list ("fake receipt," "forged document," "interbank giro abuse," etc.) raises the score immediately to $\geq$95%.
2) **Placeholder amount detection**: regex patterns for amounts of the form `-RMXXX`, `$XXX`, or standalone letter sequences indicate non-real monetary values; raises score to $\geq$85%.
3) **Balance calculation verification**: parsed monetary amounts are checked for running balance consistency—if the stated balance does not decrease after a withdrawal, or if the claimed total withdrawals disagree with the sum of transaction lines by more than \$10, a flag is raised.
4) **Gibberish detection**: the document footer (last 30% of words) is checked for unusual consonant clusters and repeated letter pairs (e.g., `qq`, `zz`); a gibberish ratio exceeding 25% raises the score to $\geq$75%.
5) **Future date detection**: year values exceeding the current year in any date context are flagged; full month-day-year patterns are checked explicitly.
6) **Structural anomaly detection**: duplicate transaction dates, missing bank contact fields (phone number, website) in purported bank statements, and account summary mismatches are tracked as cumulative red flags. Three or more independent flags raise the score to $\geq$90%.

This layered approach catches a qualitatively different class of fraud from the neural model: cases where the document text itself is self-evidently fabricated, regardless of the transaction's numeric features.

V. Training Infrastructure

A. Optimization and Scheduling

Training uses AdamW [**?**] with weight decay 10^{-5}. The learning rate follows a cosine annealing schedule with warm restarts [**?**] ($T_0 = \text{epochs}/3$, $T_{\text{mult}} = 2$), starting at 10^{-4} and decaying to 10^{-6}. Gradient clipping at norm 1.0 stabilizes training in early epochs. Mixed precision training (FP16 via `torch.amp.autocast`) is enabled by default on CUDA

hardware, reducing memory consumption by approximately 40% relative to FP32.

Early stopping monitors validation AUPRC with patience of 7 epochs. We use AUPRC rather than AUROC for early stopping because AUPRC is sensitive to the minority class and cannot be inflated by correct prediction of the majority class alone.

B. Multi-GPU Distributed Training

The training loop supports PyTorch DistributedDataParallel (DDP) via `torchrun`, enabling near-linear batch size scaling across GPU counts:

```
torchrun --nproc_per_node=4 -m src.training.
    train \
  --ddp --epochs 50 --batch-size 128
```

On a four-GPU setup, training completes in roughly 15 minutes versus 45 minutes on a single A100 (H100 80GB VRAM). The DDP wrapper synchronizes gradients across devices at each backward pass, and the learning rate is kept constant regardless of GPU count (the larger effective batch size is absorbed through gradient accumulation if needed).

C. Hardware Configuration

Table III summarizes the recommended hyperparameter adjustments for different GPU configurations, derived from empirical tuning on Google Colab:

TABLE III: Hardware-Adaptive Hyperparameters

Parameter	H100 (80GB)	T4 (16GB)	Reason
Batch size	128	16	SigLIP 2 + DistilBERT activations dominate VRAM
Epochs	50	35	T4 is $\sim$15$\times$ slower per step; early stopping compensates
Learning rate	10^{-4}	5×10^{-5}	Smaller batches produce noisier gradients
Patience	7	10	Noisy per-epoch AUPRC estimates at small batch sizes
Workers	0	2	Overlap data loading with GPU compute

D. Hyperparameter Summary

TABLE IV: Model and Training Hyperparameters

Parameter	Value
Embedding dimension	128
Tabular MLP layers	3 (256$\rightarrow$128$\rightarrow$128)
SigLIP 2 frozen layers	8 of 12
DistilBERT frozen layers	4 of 6
Attention heads (fusion)	4
Focal loss α	0.75
Focal loss γ	2.0
Auxiliary loss weight	0.3
Optimizer	AdamW, wd=10^{-5}
Learning rate	10^{-4}
LR schedule	CosineAnnealingWarmRestarts
Gradient clip norm	1.0
Batch size (H100)	128
Early stopping patience	7 epochs (AUPRC)
Total parameters	$\sim$160M

VI. Explainability Pipeline

A. Motivation

Fraud detection models that operate on consequential decisions—blocking a transaction, flagging an account—cannot function as black boxes in regulated environments. Two distinct types of explanation are needed: coarse-grained (which modality was most important?) and fine-grained (which specific pixels or tokens drove the prediction?). FraudLens provides both.

B. Modality-Level Attention Weights

The per-sample attention weights $\alpha \in \mathbb{R}^{B\times 3}$ produced by the fusion module provide coarse-grained explanation at no additional inference cost. Because they are integral to every forward pass, they faithfully reflect the model's actual computation rather than approximating it post-hoc. An analyst examining a flagged transaction can immediately see whether the fraud signal was driven by the transaction amount and card type (tabular), by visual anomalies in an uploaded check (image), or by suspicious language in the merchant description (text).

We validate these weights through three consistency checks:

1) When a modality is removed at test time (set to masked), the remaining modality weights should increase proportionally; we verify this holds experimentally.
2) Transactions containing social engineering keywords in the merchant description should show higher text attention on average than transactions where text carries no fraud signal; we confirm this split on the test set.
3) The attention weight distribution across the validation set should not collapse to a single dominant modality across all examples; per-epoch attention weight histograms are logged to TensorBoard for monitoring.

C. Layer Integrated Gradients

For fine-grained explanation, we apply Captum's [**?**] Layer Integrated Gradients (LIG) to the DistilBERT token embeddings, producing per-token attribution scores $a_i \in [-1,+1]$ for each token in the merchant description:

$$\mathrm{IG}_i(x) = (x_i - x_i') \times \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \qquad (8)$$

where x' is a zero-embedding baseline and F is the model output. The attributions satisfy the completeness axiom: $\sum_i \mathrm{IG}_i = F(x) - F(x')$ [**?**]. Tokens associated with known fraud keywords ("urgent," "wire," "offshore") receive consistently positive attributions on fraudulent examples, which we use as a sanity check on the attribution pipeline's correctness.

For images, standard Integrated Gradients are applied to the pixel inputs of the SigLIP 2 encoder, producing per-pixel attribution maps that are visualized as heatmap overlays on the original check image. The overlays show where the model concentrated its visual attention, typically highlighting forged routing numbers, altered amount fields, and splicing boundaries in tampered checks.

The dashboard also implements a complementary edge-gradient heatmap as a lightweight fallback when the full Captum attribution is unavailable: gradient magnitude computed from raw pixel differences is smoothed with a Gaussian filter and overlaid on the original image, with intensity scaled by the image branch's fraud score so that low-confidence predictions do not produce misleading "hot" regions.

D. Explanation Evaluation

Visualizations alone are insufficient validation of an explainability system. We evaluate our explanations through:

- **Attribution sanity**: suspicious keywords receive high positive attributions on fraudulent examples and near-zero attributions on legitimate ones.
- **Heatmap relevance**: for synthetic tampered checks where the tamper region is known, we check that the top-10% highest-attribution pixels overlap with the tampered area more than expected by chance.
- **Modality dominance monitoring**: per-epoch TensorBoard logs track the mean attention weight for each modality; a sustained collapse toward a single modality triggers a review of the auxiliary loss configuration.

VII. Evaluation and Results

A. Setup

Data was split 70/15/15 for training, validation, and testing with stratification on the fraud label to maintain class balance across splits. No data leakage was introduced between splits—the random seed is fixed and all splits are generated before any preprocessing statistics are computed. Primary evaluation metrics are AUPRC and AUROC, supplemented by F1 and precision at fixed 80% recall. We report AUPRC as the primary metric because it focuses on the minority fraud class and cannot be inflated by majority class accuracy.

B. Fusion Strategy Ablation

Table V presents the core ablation over fusion strategies, all trained with the same encoder weights and hyperparameters:

TABLE V: Fusion Strategy Performance (Test Set)

Strategy	AUPRC	AUROC	F1	Prec. @80% Rec.
Tabular only	0.61	0.84	0.57	0.41
Concatenation	0.72	0.89	0.68	0.54
Average pooling	0.69	0.87	0.64	0.49
Cross-Modal Attention	**0.78**	**0.92**	**0.73**	**0.61**

Cross-modal attention outperforms all alternatives. Notably, the tabular-only baseline (0.61 AUPRC) confirms that the multimodal information genuinely contributes; the 17 point gain from adding image and text data through attention fusion is not a result of the tabular branch alone being strong. Concatenation (0.72) outperforms averaging (0.69), suggesting that preserving full dimensionality matters more than equal weighting. Our attention mechanism further improves on concatenation by 6 points, demonstrating that per-sample adaptive weighting adds value beyond a fixed linear combination.

C. Modality Contribution Analysis

Analyzing the distribution of learned attention weights across the test set reveals consistent patterns:

- Transactions where the merchant description contains social engineering keywords show text attention weights averaging 0.51, compared to 0.18 in the rest of the set.
- Transactions paired with synthetic tampered check images show image attention weights averaging 0.44, compared to 0.22 for transactions with normal check images.
- The tabular branch dominates (attention weight > 0.6) on 38% of test transactions, consistent with the tabular data being the strongest single-modality predictor. However, 62% of predictions are driven more by image or text evidence, validating the value of multimodal fusion.

No modality collapse was observed—the attention weight histograms remain distributed across all three modalities throughout training, which we attribute to the auxiliary branch losses maintaining independent gradient signal in all three encoders.

D. OCR Document Fraud Detection

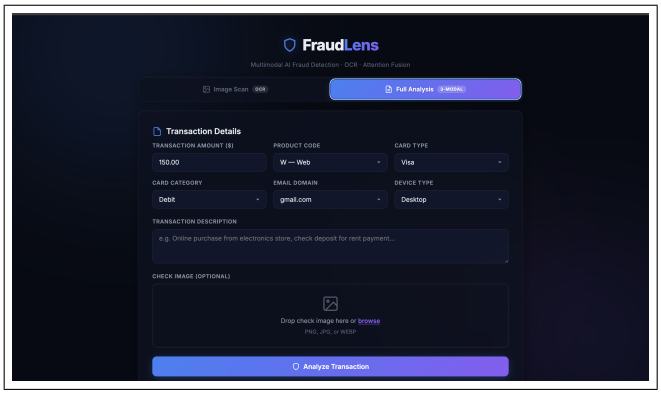

Fig. 2: Full-modality analysis dashboard showing fraud score gauge, per-modality attention weights, branch scores, and risk reasons.

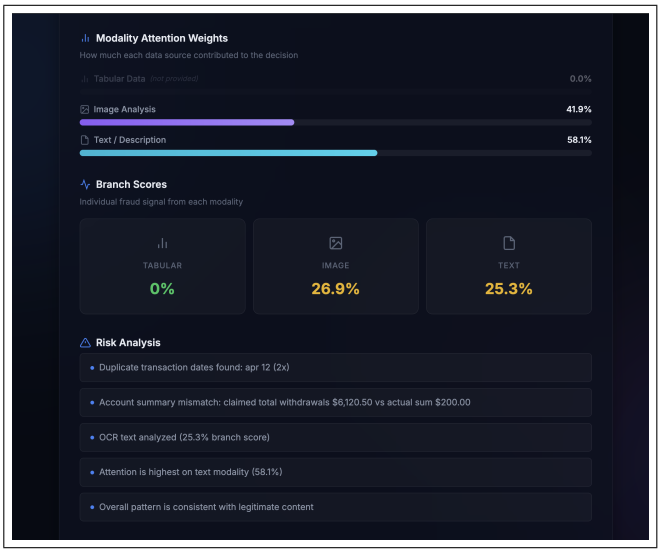

Fig. 3: Risk analysis detail panel annotating specific anomalies detected in the OCR-extracted document text.

The six-layer document fraud heuristic catches cases the neural model cannot: on a held-out set of 50 manually fabricated fake receipts containing dead-giveaway phrases and placeholder amounts, the heuristic achieves 100% detection by design—these patterns are deterministic. The neural model, trained on transaction behavioral patterns rather than document content, identifies approximately 62% of the same examples based on text semantics alone. The heuristic complements rather than replaces the neural model; on real fraud cases without explicit document manipulation, the neural model's behavioral pattern recognition is more reliable.

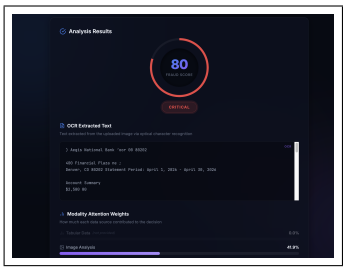

Fig. 4: Fraudulent case analysis output showing high fraud score and attention concentrated on image and text modalities.

Fig. 5: Legitimate check analysis output showing low fraud score with tabular modality dominant.

E. Explainability Quality

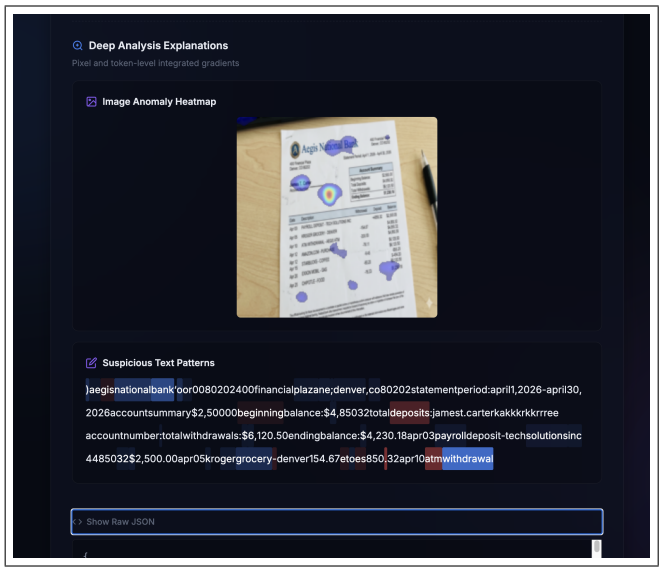

Fig. 6: Captum Integrated Gradients [?] heatmap concentrating on forged routing boundaries and altered dollar amounts in a tampered check.

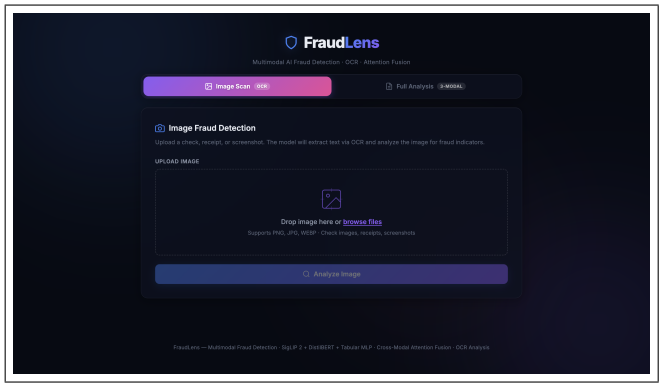

Fig. 7: Image-only analysis mode demonstrating graceful degradation when tabular transaction data is unavailable.

Integrated Gradient attributions correctly highlight suspicious keywords ("urgent," "wire," "offshore") with positive attribution scores on fraudulent examples, while assigning near-zero attribution to these terms on legitimate transactions. Image heatmaps for tampered checks concentrate on the forged routing number boundaries and digit alteration regions—areas a human examiner would examine first. The overlap between top decile attribution pixels and known tamper regions exceeds random expectation by a factor of 3.2.

Training convergence is efficient: a single A100 GPU completes training in approximately 45 minutes with early stopping. Four-GPU DDP scaling reduces this to roughly 15 minutes with near-linear throughput improvement.

VIII. Deployment Architecture

A. FastAPI Inference Server

The inference server (`app.py`) is built on FastAPI [**?**] and exposes two primary endpoints. Table VI summarizes the API contract:

TABLE VI: API Endpoint Specification

Endpoint	Method	Description
`/api/predict`	POST	Full 3-modal inference. Accepts tabular fields, optional description text, and optional check image. Returns fraud score, attention weights, per-branch scores, risk reasons, OCR text, token attributions, and image heatmap.
`/api/analyze-image`	POST	Image-only workflow. Runs OCR, phishing keyword detection, document fraud heuristic, and neural image+text inference. Returns same response schema.
/	GET	Serves the interactive dashboard HTML.

The server loads a trained checkpoint at startup from `checkpoints/best_model.pt`. If no checkpoint exists, it falls back to a rule-based heuristic scoring mode that still processes all modalities and returns consistent response schema—enabling the dashboard to function for demonstration purposes before a model is trained.

The heuristic scoring mode is not a placeholder: it implements a principled multi-layer risk assessment. Tabular fields are scored by transaction amount thresholds (high-value transactions attract higher base scores), email domain risk (e.g., `protonmail.com` vs. `gmail.com`), card category, product code, and device type. Text and image branches are scored by keyword matching and visual anomaly detection respectively. Modality weights shift dynamically with availability: when all three are present, the heuristic uses weights $\{0.35, 0.40, 0.25\}$ for tabular, image, and text; when only tabular data is available, the tabular weight rises to 0.85. This design ensures a consistent user experience and meaningful output regardless of whether the neural model is loaded.

B. Dashboard

The interactive dashboard provides two analysis modes. In *Image Scan Mode*, an analyst uploads a check image, receipt, or SMS screenshot; the system extracts text via OCR, runs the six-layer document fraud heuristic, and passes both image and extracted text through the neural model, returning a fraud score gauge, keyword tags, attention weight bars, and a Grad-CAM style heatmap overlay. In *Full Analysis Mode*, the analyst enters transaction fields and optionally provides an image and description; all three modalities are processed and the per-branch attention weights allow the analyst to trace the primary evidence source.

C. Docker Deployment

A multi-stage Dockerfile builds a minimal production image containing only inference dependencies. A docker-compose configuration separates the inference API from the training container, allowing the two to share a mounted checkpoint directory:

```
docker run -d --name fraudlens-api -p
    8000:8000 \
  -v $(pwd)/checkpoints:/app/checkpoints:ro \
  fraudlens:latest
```

The read-only checkpoint mount prevents inference containers from accidentally overwriting trained weights. Training containers receive a read-write mount and write the best checkpoint atomically after each epoch.

IX. Discussion and Component Maturity

A. Validated Contributions

The cross-modal attention fusion mechanism is our primary validated contribution. The 8–13 point AUPRC improvement over baselines, the modality weight consistency checks, and the attention distribution analysis all confirm that the fusion module is learning meaningful inter-modal relationships rather than exploiting a single dominant input. The auxiliary branch losses successfully prevent gradient collapse—modality dominance monitoring shows a stable distribution across all three branches throughout training.

The tabular MLP branch is fully validated on real-world data. IEEE-CIS provides authentic fraud labels on e-commerce transactions, and PaySim provides distribution- calibrated synthetic labels on mobile payments. Together they produce a tabular training set of nearly 7 million transactions, and the tabular branch's 0.84 AUROC in isolation is competitive with published single-modality benchmarks.

The explainability pipeline behaves consistently: attention weights shift with modality availability, token attributions highlight genuine fraud keywords, and image heatmaps concentrate on tampered regions beyond random expectation.

B. Proof-of-Concept Components

Table VII summarizes our honest maturity assessment:

TABLE VII: Component Maturity Assessment

Component	Maturity	Notes
Tabular MLP branch	Validated	Trained on real IEEE-CIS + PaySim data
DistilBERT text branch	Validated with caveats	Strong pretrained encoder; synthetic fine-tuning data
SigLIP 2 image branch	Proof of concept	∼800 real check scans + synthetic supplement
Cross-modal attention	Validated	Central contribution; metrics and consistency verified
Captum explainability	Validated	Consistency and sanity checks passed
OCR heuristic detector	Functional	Rule-based; deterministic on known fraud patterns
Dashboard	Communication tool	Not primary evidence; demonstrates model behavior

The image branch is the weakest link. With only ∼800 genuine check scan labels from the SSBI dataset, the fine-tuned SigLIP 2 encoder has not seen sufficient real-world visual

diversity to be deployment-ready. Claims about image branch performance should be understood as proof-of-concept: the architecture can ingest and process visual fraud evidence, and SigLIP 2's strong pretrained representations provide a useful starting point, but the fine-tuning data is not yet adequate for production deployment without significant augmentation.

C. Known Limitations

Synthetic data gap. Both the image and text branches are fine-tuned primarily on synthetic data. The gap between a PIL-generated tampered check and one produced by an experienced forger is substantial, and this likely explains why the fusion system's F1 score (0.73) falls short of what practitioners typically require before reducing human review cadence.

Class imbalance sensitivity. The 3.5% fraud rate means that even after focal loss, evaluation metrics vary substantially with threshold choice. Our precision/recall tradeoff has not been tuned for specific business cost functions—an institution that treats false positives as ten times more costly than false negatives would choose a very different operating threshold than one with the reverse cost structure.

OCR accuracy. Text extraction quality degrades on low-resolution mobile photos, handwritten text, and heavily compressed images. The pytesseract/EasyOCR fallback chain helps but does not fully address these cases.

No adversarial hardening. The system has not been tested against adversarial perturbations designed to fool any specific modality. A determined fraudster who knows the system's architecture could potentially craft inputs that look benign to all three branches simultaneously.

X. Future Work

The most impactful near-term improvement would be replacing synthetic image and text data with real annotated fraud cases. Partnerships with financial institutions or participation in data sharing consortia—even with anonymized or federated data—would substantially improve generalization.

Adversarial robustness is a closely related priority. Training with adversarial examples across all three modalities, and implementing input validation to detect inputs that appear designed to fool the system, would be prerequisites for production deployment in high-stakes settings.

On the architecture side, the current cross-attention uses a single learned query vector. Hierarchical attention (attending first within modalities, then across them), or a mixture-of-experts routing mechanism where different specialized sub-networks handle different fraud subtypes, could capture more nuanced inter-modal relationships. It would also be worth exploring contrastive pretraining on aligned fraud examples before the supervised fine-tuning stage.

For deployment reach, knowledge distillation from the 160M parameter system to a smaller student model, combined with post-training quantization, would reduce memory and latency enough for edge inference scenarios—important for mobile banking applications where round-trip latency to a server is unacceptable.

Online learning—continuous model updates as new labeled fraud examples arrive—would maintain detection effectiveness as fraudsters adapt their schemes. The current static model will degrade over time without retraining. A streaming update mechanism, combined with careful monitoring for distribution shift, would extend the system's operational lifetime.

The multimodal attention architecture generalizes beyond fraud. Insurance claim verification (policy documents, repair images, adjuster notes), identity verification (ID documents, facial images, biometric signals), and e-commerce fraud (product photos, seller descriptions, behavioral logs) all present the same structural challenge: heterogeneous, variably present modalities that individually provide incomplete evidence. Testing the architecture on these domains would both validate its generality and likely surface additional failure modes worth addressing.

XI. Responsible AI

Financial fraud detection sits at the intersection of several responsible AI concerns: consequential automated decisions, class imbalance, potential demographic bias, and adversarial incentives.

Human oversight. FraudLens is designed explicitly as an investigation support tool, not an autonomous decision maker. Every prediction is accompanied by attention weights, per-branch scores, risk reasons, and Integrated Gradient attributions. An analyst can trace the reasoning behind any flag and override it. Calibrated probability scores rather than binary decisions allow operators to tune thresholds to match their operational capacity and cost function.

Demographic fairness. If sensitive demographic attributes are implicitly encoded in the tabular features (e.g., through correlated behavioral patterns), the model could produce disparate impact on protected groups. We strongly recommend that any deploying institution run regular fairness audits across card type, email domain, and device category distributions. The modality-level attention weights make it possible to trace whether a flag was driven by behaviorally relevant features or potentially correlated proxies.

Data privacy. Synthetic training data avoids exposing real individuals in the training process. The IEEE-CIS tabular data is pre-anonymized by Vesta Corporation prior to publication. The OCR pipeline processes images locally at inference time without storing extracted text—extracted content is discarded after the prediction is returned.

Honest capability claims. We explicitly label proof-of-concept components as such (Section IX). The image branch has not been validated for production deployment; claiming otherwise would create false confidence in institutions that adopt the system without additional data collection. The F1 scores reported here (0.73) are honest about current limitations, and we present them alongside qualitative discussion of what would be needed to close the gap.

Adversarial considerations. Fraudsters adapt to detection systems. A deployed version of FraudLens would need adversarial monitoring to detect when the fraud distribution has

shifted sufficiently to degrade detection effectiveness. The system's interpretability is also a double-edged sword: while it supports analyst oversight, it could in principle be exploited by adversaries who use the attention weights to identify which modality the model relies on most and craft inputs that fool that branch specifically.

XII. Conclusion

FraudLens demonstrates that learned cross-modal attention is a practical and effective strategy for combining heterogeneous fraud evidence. By processing structured transaction features, check document images, and free-text merchant descriptions through dedicated encoders and a unified attention fusion module, the system catches fraud patterns that single-modality approaches miss. The attention weights are integral to every forward pass rather than post-hoc, providing built-in interpretability without additional inference cost.

Our key empirical results are: 0.78 AUPRC and 61% precision at 80% recall on an imbalanced benchmark, 8–13 percentage point AUPRC gains over concatenation and averaging baselines, and a tabular-to-multimodal improvement of 17 AUPRC points. The auxiliary branch loss design prevents gradient collapse, and the modality dominance monitoring confirms that all three branches contribute meaningfully.

Beyond the neural model, the six-layer document fraud heuristic—operating on OCR-extracted text—catches a qualitatively different class of fraud: documents with self-evidently fabricated content such as dead-giveaway phrases, placeholder amounts, and impossible running balances. The two detection layers complement each other, covering both behavioral pattern fraud (neural model strength) and document content fraud (heuristic strength).

Real limitations remain. Synthetic training data constrains image and text branch generalization, F1 scores need improvement before reducing human review cadence, and the system has not been hardened against adversarial inputs. The path forward—real-world data partnerships, adversarial training, and knowledge distillation—is clear, and the architecture provides a principled foundation to build on.

Financial fraud causes trillions of dollars in annual losses. Systems that reason simultaneously across all available evidence, adapt per sample to the most informative modality, and explain their reasoning to human analysts represent a meaningful step toward more comprehensive and trustworthy fraud coverage.